

SIH26099 • ALCHEMY ARCHITECTS

CPCL / MoPNG

ENTERPRISE V2.4

MATRA-X

Autonomous Material Master Harmonization Engine with Deterministic Safety Firewalls & Cross-CPSE Knowledge Graphs.



6-Stage Pipeline

Multi-source Ingest



Hybrid AI Engine

Vector + Cross-Encoder



False-Merge Guard

Deterministic Firewall

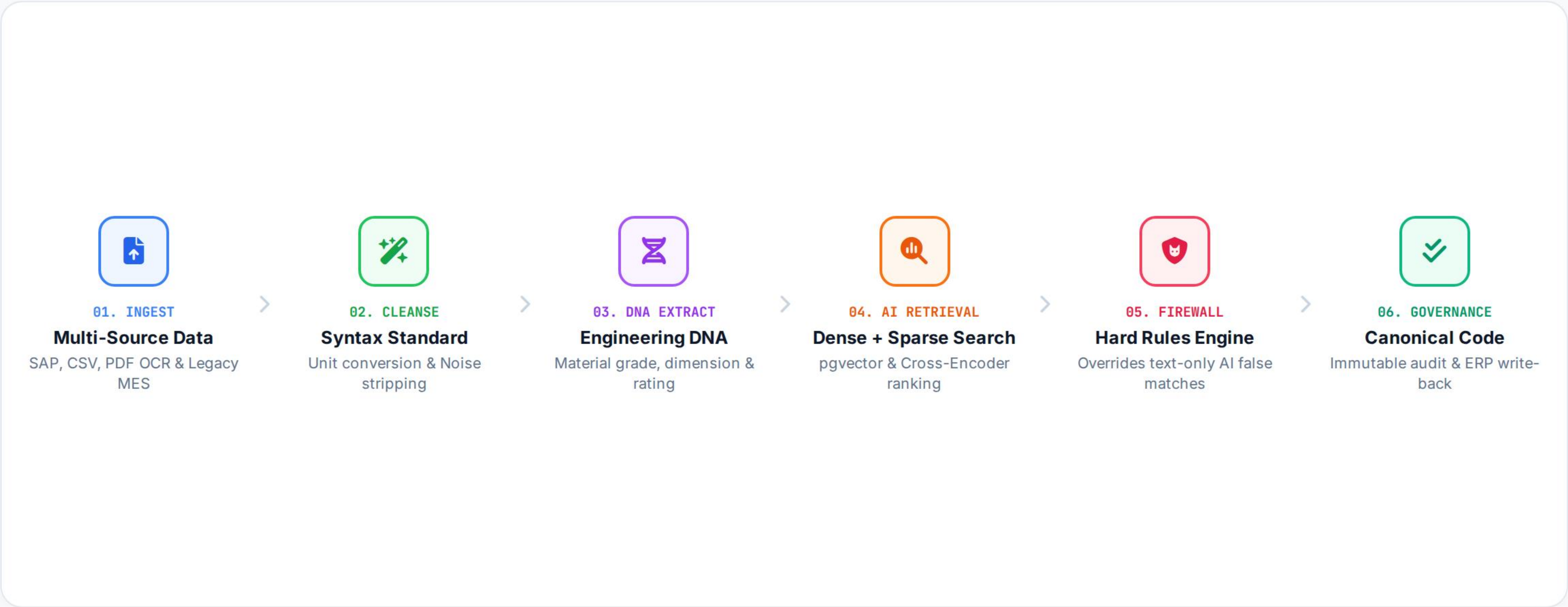


KnowLedge Mesh

Neo4j Lineage Graph

End-to-End Autonomous Harmonization Flowchart

Data Ingestion → Governed Master ID



Auto Execution Mode: High confidence matches (> 92% + 0 firewall flags) routed directly to auto-creation.



Human-in-the-Loop Mode: Borderline scores (70–92%) or material property conflicts flagged for expert sign-off.

Hybrid Vector & Semantic Scoring Mechanics

$$\text{Score} = \alpha \cdot S_{\text{dense}} + \beta \cdot S_{\text{sparse}} + \gamma \cdot S_{\text{struct}}$$

STEP 1 • VECTOR RETRIEVAL



Dense Embedding Top-K

Domain-fine-tuned Sentence-BERT translates raw text into 768-dim embeddings in pgvector space.

ALGORITHM

Cosine Distance / HNSW Index

STEP 2 • PROPERTY MATCH

Sparse Spec Extraction

Deterministic regex parser extracts discrete engineering keys (Grade, Rating, Size, Standard).

FEATURE MATRIX

Structured Field Comparison

STEP 3 • RE-RANKING



Cross-Encoder Deep Scoring

Pairwise transformer re-ranker validates full textual context against Top-50 candidate items.

OUTPUT CONFIDENCE

Normalized Probability (0 - 1.00)

LIVE MODEL EXECUTION

Input: "GATE VALVE 2 INCH CLASS 150 FLANGED A105" → Matched Master ID: VAL-GT-150-002 (Confidence: 0.968)

False-Merge Safety Firewall Architecture

LLM / Vector Search Risk

THE HARD TRAP

Pure AI embeddings assign > 95% similarity to items with identical textual structure but incompatible metallurgy or ratings.

ILLUSTRATIVE DANGER CASE

Item A: STAINLESS STEEL 304 BOLT M16

Item B: STAINLESS STEEL 316 BOLT M16

⚠️ AI Text Similarity: 98% (High risk false merge!)

Failure Effect: Plant Shutdown / Safety Hazard

MATRA-X Firewall Rules Engine

DETERMINISTIC GUARD



Rule 1: Material Grade Mismatch

Hard block when SS304 vs SS316 vs CS grade conflict occurs.



Rule 2: Pressure & Temperature Rating

Class 150 vs Class 300 vs Class 600 strict boundary filter.



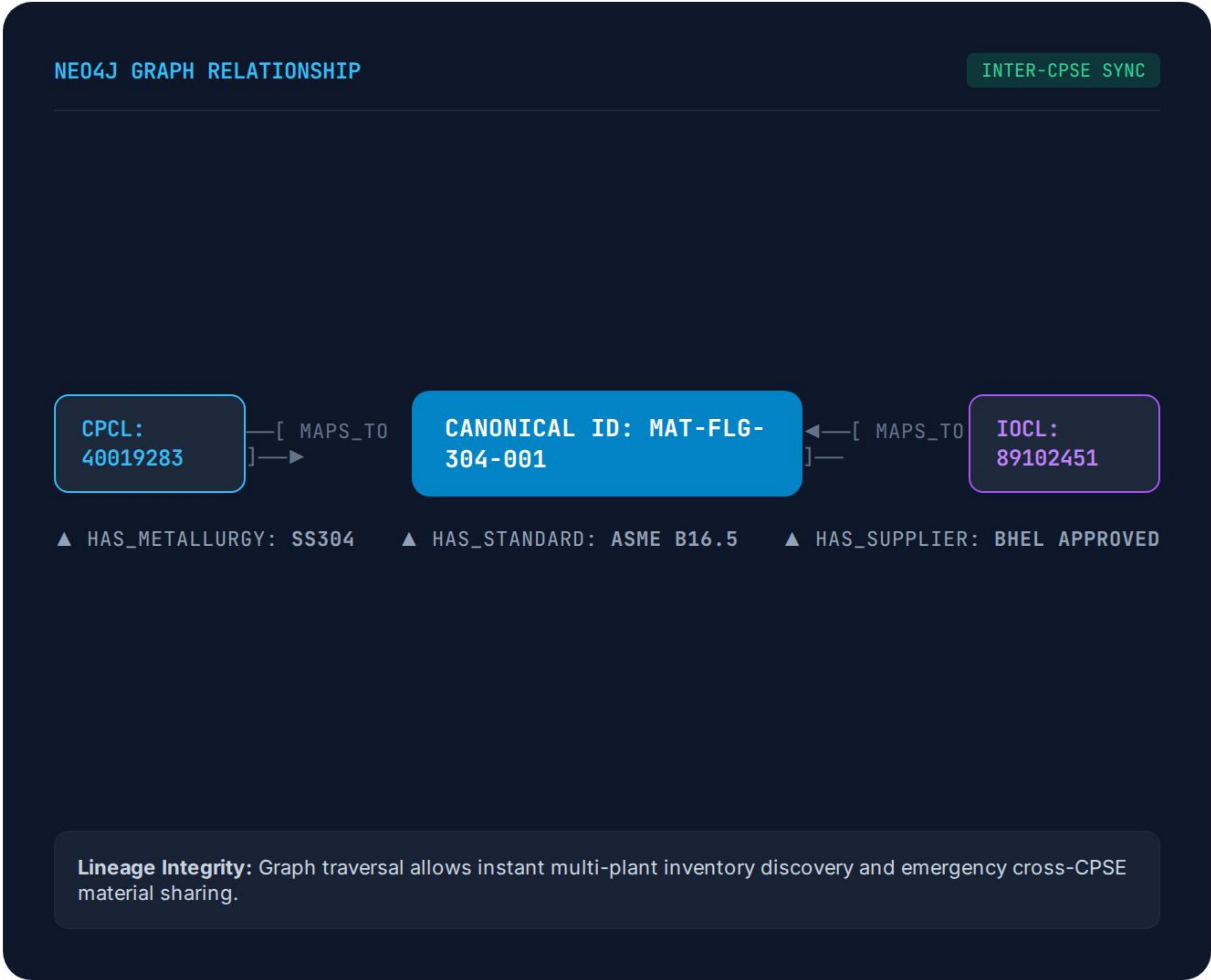
Rule 3: Dimension & Unit Variance

Prevents 2" metric vs 50mm conversion ambiguity mismatches.

FIREWALL OUTCOME

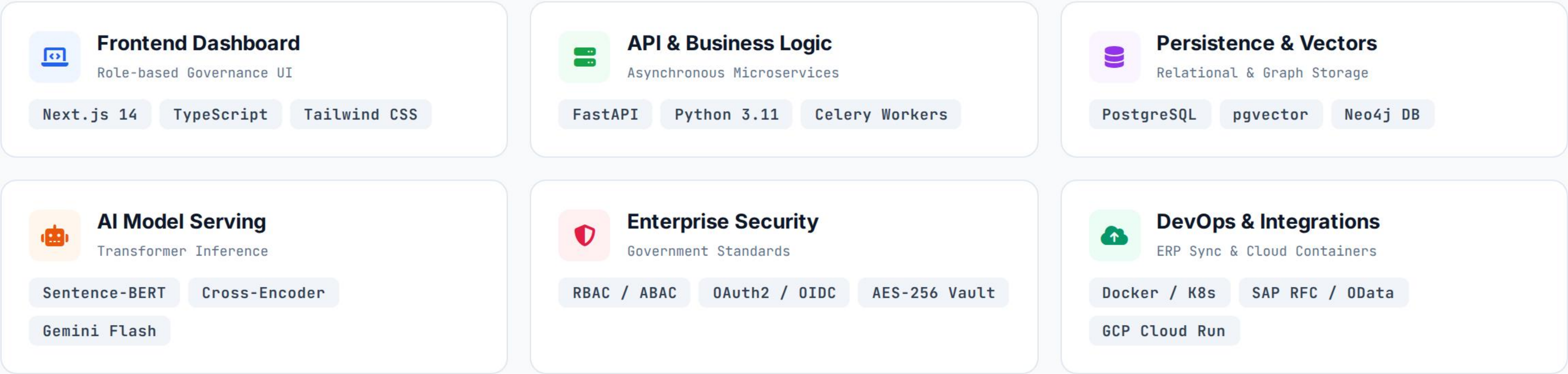
100% FALSE-MERGE PREVENTION

Neo4j Knowledge Mesh & Material Graph

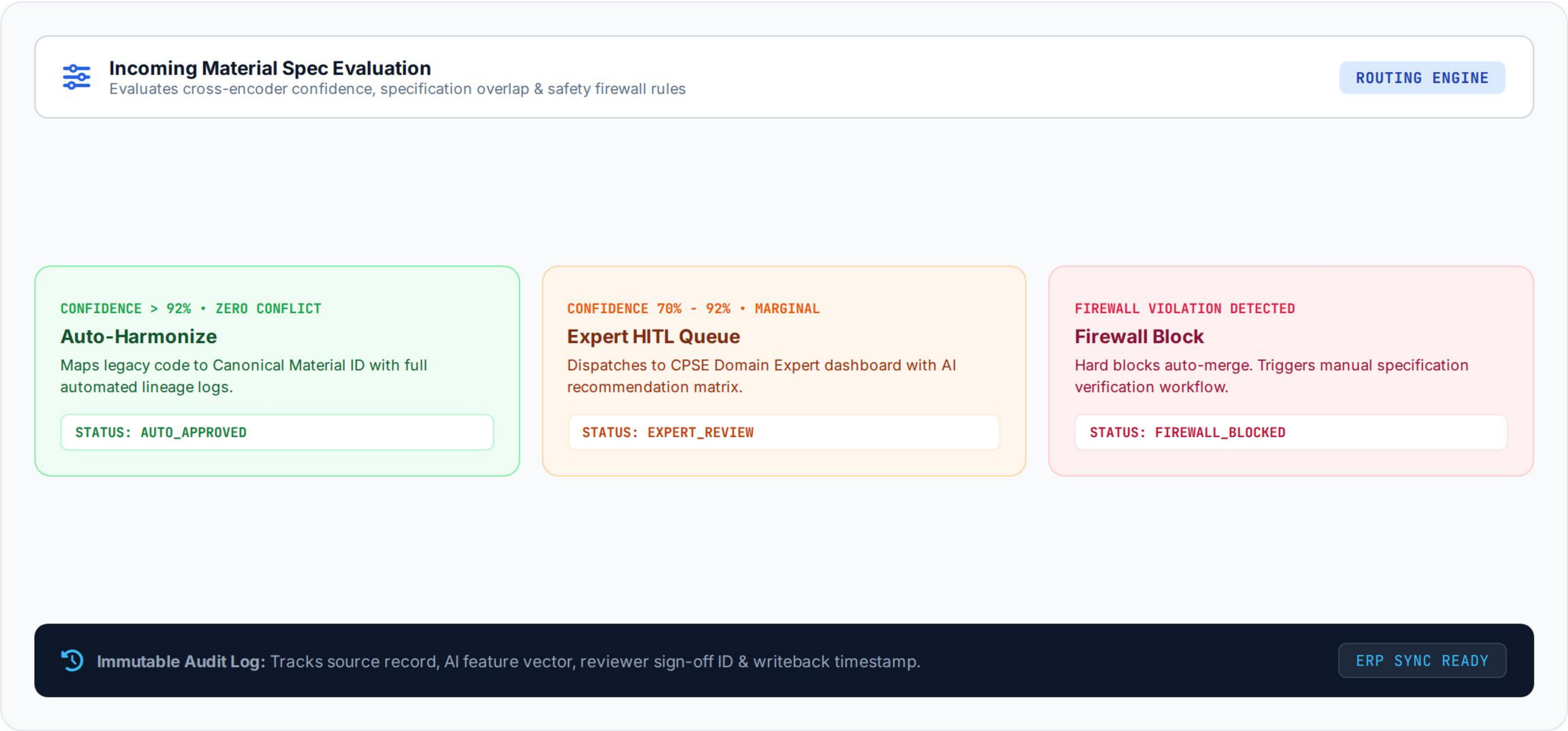


- Cross-CPSE Interoperability**
Unifies duplicate codes across CPCL, IOCL, ONGC, and HPCL into standard national catalog.
- Full Provenance & Lineage**
Every node tracks creation timestamp, user overrides, supplier history, and audit trail.
- Dead Stock Liquidation**
Identifies idle inventory across nearby refineries to prevent duplicate procurement orders.

Production Microservices & Cloud Stack



Multi-Tier Decision Matrix & Human-in-the-Loop Queue





ONE NATION • ONE MATERIAL CODE

Empowering Indian CPSEs with AI-Driven Data Sovereignty, Zero False Merges, and Unified Procurement.

85%+

Automated Code Mapping

0%

Safety False-Merge Rate

₹100Cr+

Dead-Stock Optimization Potential

TEAM NAME

Alchemy Architects

PROBLEM STATEMENT

SIH26099 (CPCL / MoPNG)

MATRA-X System Overview Complete